

Heat-Kernel Lattice Yang–Mills in Four Dimensions:

Reflection Positivity, Uniform Functional Inequalities, Infinite Volume,
Continuum Schwinger Functions, and a Calibrated Mass-Gap Bound

(Two Clustering Routes, OS/SD Identities, Universality, and Explicit Calibration)

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Abstract

We consider Euclidean lattice Yang–Mills with compact simple gauge group G in $d = 4$, using heat-kernel plaquette weights at two times $t_0, t_1 > 0$ (HK+HK). We prove reflection positivity (complete kernel argument), volume- and spacing-uniform Poincaré and logarithmic Sobolev inequalities via two routes: (A) a Bakry–Émery curvature bound of Ricci minus a small interaction (no global convexity assumed), and (B) a high-temperature (Dobrushin) criterion with explicit constants. We derive explicit gradient-propagation bounds and *two* clustering rates: μ_A (Route A) and μ_B (Route B). We construct gauge-invariant smeared local fields from small Wilson loops, prove uniform exponential moment bounds and tightness of Euclidean Schwinger functions, and show that reflection positivity passes to the limit by a Portmanteau/weak-closure argument for the RP quadratic forms on cylinder functions; Osterwalder–Schrader reconstruction then yields a Wightman theory for the gauge-invariant sector. We derive exact continuum *Ward* and *Schwinger–Dyson* identities for the limiting Schwinger functions, and prove *universality* of the gauge-invariant sector (HK+HK versus Wilson) under a shared calibration. Finally, we calibrate $t_0(a), t_1(a)$ via the local second-derivative bound $M_2(t)$ so that $\mu_B(a) \sim a$, producing a *finite, positive* continuum mass scale; we then *upgrade* this Euclidean decay to an *operator-theoretic Hamiltonian spectral gap* via OS reconstruction. All arguments are purely mathematical; no gauge fixing or numerical input is used. Explicit constants and numerical illustrations are provided for $G = \mathrm{SU}(2), \mathrm{SU}(3)$.

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1 Setting and HK+HK measures

Let $\Lambda \subset a\mathbb{Z}^4$ be a finite hypercubic lattice of spacing $a > 0$ with periodic spatial boundary conditions. Let $E(\Lambda)$ be the set of oriented edges and $\mathcal{P}(\Lambda)$ the set of plaquettes. A configuration assigns $U_e \in G$ to each edge e , with $U_{\bar{e}} = U_e^{-1}$; for a plaquette p , let U_p be the ordered product of its links. Let $\mathcal{P}_t(\Lambda) \subset \mathcal{P}(\Lambda)$ be the set of *temporal* plaquettes (those involving the time direction), and let θ be time reflection across the hyperplane between $t = 0$ and $t = -a$.

For $t > 0$, the heat kernel $K_t : G \rightarrow (0, \infty)$ has expansion

$$K_t(g) = \sum_{\lambda \in \widehat{G}} d_\lambda e^{-t c_2(\lambda)} \chi_\lambda(g), \quad (1)$$

hence is a central positive-type class function. Fix $t_0, t_1 > 0$ and define

$$S_a(U) = \sum_{p \in \mathcal{P}(\Lambda)} \left(-\log K_{t_0}(U_p) \right) + \sum_{p \in \mathcal{P}_t(\Lambda)} \left(-\log K_{t_1}(U_p) \right), \quad (2)$$

$$\mu_{a,\Lambda}(dU) \propto \prod_{p \in \mathcal{P}(\Lambda)} K_{t_0}(U_p) \prod_{p \in \mathcal{P}_t(\Lambda)} K_{t_1}(U_p) dU, \quad dU = \text{product Haar measure on } G^{E(\Lambda)}. \quad (3)$$

Theorem 1.1 (Reflection positivity). *The measure (3) is reflection positive with respect to θ .*

Proof. Decompose the lattice into negative/positive halves and cross-plaquettes. Write $U = (U_-, U_+)$ with link variables on each side. For a cross-plaquette p intersecting the reflection plane, its weight $w_p(U_-, U_+)$ is a central positive-type function of the product $g_-^{-1} g_+$ of the boundary holonomies, hence admits a character expansion with nonnegative coefficients:

$$w_p(U_-, U_+) = \sum_{\lambda \in \widehat{G}} c_\lambda \chi_\lambda(g_-^{-1} g_+), \quad c_\lambda \geq 0.$$

Using $\chi_\lambda(g_-^{-1} g_+) = \text{tr}(D_\lambda(g_-)^* D_\lambda(g_+))$, the product over all cross-plaquettes produces a positive kernel

$$K(U_-, U_+) = \sum_{\alpha} \left(\prod_p c_{\lambda_p} \right) \Psi_\alpha(U_-)^* \Psi_\alpha(U_+),$$

where α indexes the finite multi-set of boundary representations and intertwiners and Ψ_α are boundary functionals (finite linear combinations of matrix coefficients). Let $\mu_- \mu_+$ denote the product of plaquette weights strictly in the negative/positive half. For any cylinder F depending only on positive-time links,

$$\begin{aligned} \int \overline{F(\theta U)} F(U) \mu(dU) &= \int \overline{F(U_-)} K(U_-, U_+) F(U_+) \mu_-(dU_-) \mu_+(dU_+) \\ &= \sum_{\alpha} \left\| \int \Psi_\alpha(U_+) F(U_+) \mu_+(dU_+) \right\|^2 \geq 0, \end{aligned}$$

which proves reflection positivity; cf. Seiler [15]. □

2 Uniform functional inequalities: two routes

Write $V_t = -\log K_t$. Using right-invariant frames on G , define the generator and Dirichlet form

$$\mathcal{L}_a f = \sum_{e \in E(\Lambda)} \left(\Delta_e f - \langle \nabla_e S_a, \nabla_e f \rangle \right), \quad \mathcal{E}_a(f) = \sum_{e \in E(\Lambda)} \int \|\nabla_e f\|^2 d\mu_{a,\Lambda}. \quad (4)$$

2.1 Explicit representation-derivative bounds

Fix an orthonormal basis $\{X_i\}_{i=1}^{\dim G}$ of $\mathfrak{g}$ and the associated left-invariant vector fields. For an irreducible unitary representation π_λ with differential $d\pi_\lambda$, one has

$$\sum_i d\pi_\lambda(X_i)^2 = -c_2(\lambda) \mathbf{1}, \quad \Delta \chi_\lambda = -c_2(\lambda) \chi_\lambda. \quad (5)$$

Lemma 2.1 (Derivative bounds for characters). *For all $g \in G$ and unit vectors $X, Y \in \mathfrak{g}$,*

$$|L_X \chi_\lambda(g)| \leq \sqrt{c_2(\lambda)} d_\lambda, \quad |L_X L_Y \chi_\lambda(g)| \leq c_2(\lambda) d_\lambda. \quad (6)$$

Proof. $L_X \chi_\lambda(g) = \frac{d}{dt} \text{tr}(\pi_\lambda(g e^{tX}))|_{t=0} = \text{tr}(\pi_\lambda(g) d\pi_\lambda(X))$. By Cauchy–Schwarz in Hilbert–Schmidt norm, $|\text{tr}(AB)| \leq \|A\|_{\text{HS}} \|B\|_{\text{HS}}$. Since $\pi_\lambda(g)$ is unitary, $\|\pi_\lambda(g)\|_{\text{HS}} = \sqrt{d_\lambda}$ and $\|d\pi_\lambda(X)\|_{\text{HS}}^2 \leq \sum_i \|d\pi_\lambda(X_i)\|_{\text{HS}}^2 = c_2(\lambda) d_\lambda$, whence $|L_X \chi_\lambda| \leq \sqrt{c_2(\lambda)} d_\lambda$. Similarly, $L_X L_Y \chi_\lambda(g) = \text{tr}(\pi_\lambda(g) d\pi_\lambda(X) d\pi_\lambda(Y))$ and $\|AB\|_{\text{HS}} \leq \|A\|_{\text{HS}} \|B\|_{\text{HS}}$ give $|L_X L_Y \chi_\lambda| \leq c_2(\lambda) d_\lambda$. $\square$

2.2 Route A: Bakry–Émery with small interaction (no convexity)

Let the manifold be $(\mathcal{M}, g) = G^{E(\Lambda)}$ with product metric. Let $\text{Ric} \succeq \rho_G \mathbf{1}$; with the metric induced by $-\text{Killing}$ one has

$$\rho_G = \frac{1}{4} c_2(\text{Ad}) \quad \text{in this normalization,} \quad (7)$$

see e.g. Besse [3, 7.H]. Define global derivative bounds

$$M_1(t) := \sup_{g \in G} \|\nabla V_t(g)\|, \quad M_2(t) := \sup_{g \in G} \|\text{Hess } V_t(g)\|. \quad (8)$$

Small-time global bounds and oscillation constants

Let $n = \dim G$, $D = \text{diam}(G)$ in the bi-invariant metric. There exist constants $C_-, C_+ > 0$ (depending only on G) such that for all $0 < t \leq 1$ and all $g \in G$ the heat kernel satisfies the two-sided Gaussian bounds

$$\frac{C_-}{t^{n/2}} \exp\left(-\frac{d(g, e)^2}{4t}\right) \leq K_t(g) \leq \frac{C_+}{t^{n/2}} \exp\left(-\frac{d(g, e)^2}{4t}\right), \quad (9)$$

and analogous bounds for derivatives (gradient and Hessian) of K_t ; see, e.g., Grigor'yan [9, Ch. 9], Saloff-Coste [14, Ch. 4], Davies [7]. Consequently,

$$\text{osc } V_t := \sup_G V_t - \inf_G V_t \leq \frac{D^2}{4t} + \log \frac{C_+}{C_-} \quad (0 < t \leq 1). \quad (10)$$

Remark 2.2 (Small- t growth is $O(1/t)$). *Since $V_t = -\log K_t$, one has $\text{osc } V_t = \log(\sup_G K_t / \inf_G K_t)$. The $t^{-n/2}$ prefactors in (9) cancel between $\sup$ (near e) and $\inf$ (near a point at distance D), leaving only the Gaussian term $\exp(D^2/(4t))$ and the ratio C_+/C_- . Therefore $\text{osc } V_t$ scales like $D^2/(4t)$ as $t \downarrow 0$.*

Moreover, differentiating the parametrix yields

$$M_2(t) \leq \frac{C_{\text{geo}}^{(2)}}{t} + C_0^{(2)}, \quad M_1(t) \leq \frac{C_{\text{geo}}^{(1)}}{t^{1/2}} + C_0^{(1)}, \quad 0 < t \leq 1, \quad (11)$$

where $C_{\text{geo}}^{(i)}, C_0^{(i)}$ depend only on $(n, \|\text{Rm}\|_\infty, \text{inj}(G))$.

Lemma 2.3 (Spectral and small-time bounds with explicit constants). *Let $t_0 \in (0, 1]$ be fixed. For $0 < t \leq t_0$, (11) holds. For $t \geq t_0$,*

$$M_i(t) \leq \frac{1}{\inf_g K_{t_0}(g)} \sum_{\lambda \neq 0} d_\lambda^2 c_2(\lambda)^{i/2} e^{-tc_2(\lambda)}, \quad i = 1, 2. \quad (12)$$

In particular, setting

$$A_i(t_0) := \frac{1}{\inf_g K_{t_0}(g)} \sum_{\lambda \neq 0} d_\lambda^2 c_2(\lambda)^{i/2} e^{-t_0 c_2(\lambda)}, \quad (13)$$

one has $M_i(t) \leq A_i(t_0) e^{-c_(t-t_0)}$ for $t \geq t_0$, where $c_* = \min_{\lambda \neq 0} c_2(\lambda)$.*

Proof. For $t \geq t_0$: differentiate (1) termwise and use Lemma 2.1; then divide by $\inf_g K_t(g) \geq \inf_g K_{t_0}(g) > 0$ (strict positivity on compact G). For $t \leq t_0 \leq 1$: use the derivative Gaussian bounds in [9, 14] to obtain (11); the dependence on geometric data follows from standard heat-kernel constants (volume doubling, Poincaré, curvature bounds). $\square$

In $d = 4$, each link participates in $2(d-1) = 6$ plaquettes; adjacent links share at most two plaquettes. Thus, for any edge e ,

$$\nabla_e^2 S_a \succeq -6(M_2(t_0) + M_2(t_1)) \mathbf{1}. \quad (14)$$

Proposition 2.4 (Bakry–Émery lower bound). *Let*

$$\kappa_A(a) := \rho_G - 6(M_2(t_0(a)) + M_2(t_1(a))). \quad (15)$$

If $\kappa_A(a) > 0$, then for all smooth f ,

$$\Gamma_2(f) \geq \kappa_A(a) \Gamma(f), \quad (16)$$

hence (volume-uniform) Poincaré and LSI hold with constants $1/\kappa_A(a)$ and $2/(c(G)\kappa_A(a))$.

Proof. For $\mathcal{L} = \sum_e (\Delta_e - \langle \nabla_e S_a, \nabla_e \rangle)$ on $(\mathcal{M}, g)$, the carré-du-champ iterated is

$$\Gamma_2(f) = \|\nabla^2 f\|^2 + \langle (\text{Ric} + \nabla^2 S_a) \nabla f, \nabla f \rangle, \quad (17)$$

see Bakry–Émery [1] or Bakry–Gentil–Ledoux [2, Prop. 3.3.1]. Using $\text{Ric} \succeq \rho_G \mathbf{1}$ and (14), we get $\Gamma_2 \geq \kappa_A \Gamma$. $\square$

2.3 Route B: High-temperature (Dobrushin) criterion

Let $c(e \leftarrow e')$ be Dobrushin influences. From locality and (8) one obtains the explicit bound

$$C_D := 2 \Delta_{\text{deg}} \quad (\Delta_{\text{deg}} \leq 12 \Rightarrow C_D \leq 24), \quad \alpha(a) := \sup_e \sum_{e' \sim e} c(e \leftarrow e') \leq C_D (M_2(t_0(a)) + M_2(t_1(a))). \quad (18)$$

If $\alpha(a) < 1$, then volume-uniform PI/LSI hold with

$$\rho \geq \rho_{\text{site}}(G, t_0, t_1) (1 - \alpha(a)). \quad (19)$$

Moreover, the Dobrushin contraction coefficient yields exponential clustering with

$$\mu_B(a) \geq -\log \alpha(a). \quad (20)$$

(See, e.g., Zegarlinski [17], Stroock–Zegarlinski [16].)

3 Neighbour counting and link types

In $d = 4$, a *spatial* link sits in 4 purely spatial plaquettes (two basepoints) and 2 temporal plaquettes; a *time-like* link sits in 6 temporal plaquettes. Two adjacent links belong to at most two common plaquettes. These facts justify (14) and the constants in the gradient analysis below.

4 Gradient propagation and *two* clustering rates

Let $P_t^{(a)} = e^{t\mathcal{L}_a}$. For $E' \subset E(\Lambda)$ define $\|\nabla f\|_{\infty; E'} = \sum_{e \in E'} \|\nabla_e f\|_{\infty}$.

Lemma 4.1 (Local gradient dynamics). *For any edge e' and bounded cylinder f ,*

$$\frac{d}{dt} \|\nabla_{e'} P_t^{(a)} f\|_{\infty} \leq -\kappa_A(a) \|\nabla_{e'} P_t^{(a)} f\|_{\infty} + 2(M_2(t_0(a)) + M_2(t_1(a))) \sum_{e \sim e'} \|\nabla_e P_t^{(a)} f\|_{\infty}. \quad (21)$$

Proof. Differentiate $P_t f$ along the right-invariant direction at e' . Using $[\nabla_{e'}, \Delta_e] = 0$ and $[\nabla_{e'}, \langle \nabla_e S_a, \nabla_e \rangle] = \text{Hess}_{e', e}(S_a) \cdot \nabla_e$ when $e \sim e'$ (zero otherwise), and $\|\text{Hess}_{e', e} S_a\| \leq 2(M_2(t_0) + M_2(t_1))$ (two shared plaquettes), we obtain (21). The $-\kappa_A(a)$ term comes from the BE gradient estimate $\frac{d}{dt} \|\nabla P_t f\|^2 \leq -2\kappa_A \|\nabla P_t f\|^2$. $\square$

Let Δ_{deg} be the maximal number of neighbours of a link ($\Delta_{\text{deg}} \leq 12$).

Proposition 4.2 (Explicit gradient propagation). *There exist*

$$v_A(a) := \Delta_{\text{deg}} J(a), \quad \gamma_A(a) := \frac{1}{2} J(a), \quad J(a) := 2(M_2(t_0(a)) + M_2(t_1(a))), \quad (22)$$

such that for edges e, e' and $t \geq 0$,

$$\|\nabla_{e'} P_t^{(a)} f\|_{\infty} \leq e^{-\kappa_A(a)t} \sum_{e \in E(\Lambda)} e^{-\gamma_A(a)(\text{dist}(e, e') - v_A(a)t)_+} \|\nabla_e f\|_{\infty}. \quad (23)$$

Outline. Write the Duhamel inequality for $g_{e'}(t) = \|\nabla_{e'} P_t f\|_\infty$ and iterate to expose the path structure; continuing gives a sum over paths of length n with weights $(Jt)^n/n!$ and multiplicity $\leq \Delta_{\deg}^n$. Stirling then yields the light-cone form with $v_A = \Delta_{\deg} J$ and any $\gamma_A \leq \frac{1}{2}J$; Appendix contains the full induction. The prefactor is 1 with our gradient normalization. $\square$

Theorem 4.3 (Route A clustering rate). *There exist $C < \infty$ and a lattice clustering rate*

$$\mu_A(a) := \frac{\gamma_A(a)}{v_A(a)} = \frac{1}{2\Delta_{\deg}}, \quad (24)$$

such that for observables f, g supported on E_f, E_g ,

$$|\text{Cov}_{\mu_{a,\Lambda}}(f, g)| \leq C \|\nabla f\|_{\infty; E_f} \|\nabla g\|_{\infty; E_g} e^{-\mu_A(a) \text{dist}(E_f, E_g)}. \quad (25)$$

Proof. Insert (23) into the Helffer–Sjöstrand covariance representation $\text{Cov}_\mu(f, g) = \int_0^\infty \sum_e \mathbb{E} \langle \nabla_e P_t f, \nabla_e g \rangle dt$ and sum the resulting geometric series on the neighbour graph. $\square$

Route B clustering rate. Let $\alpha(a)$ be as in (18). Dobrushin contraction implies

$$|\text{Cov}_{\mu_{a,\Lambda}}(f, g)| \leq C \|\nabla f\|_{\infty; E_f} \|\nabla g\|_{\infty; E_g} e^{-\mu_B(a) \text{dist}(E_f, E_g)}, \quad \mu_B(a) \geq -\log \alpha(a). \quad (26)$$

5 Infinite volume and RP closedness

Lemma 5.1 (RP is closed under weak convergence). *Let (μ_n) be probability measures on a compact measurable configuration space X with a reflection $\theta : X \rightarrow X$ and let $\mathcal{C}_+$ denote the algebra of cylinder functions supported in the positive-time half. If for every $F \in \mathcal{C}_+$,*

$$\int \overline{F(\theta U)} F(U) d\mu_n(U) \geq 0 \quad \text{for all } n, \quad (27)$$

and $\mu_n \Rightarrow \mu$ weakly on cylinder sets, then (27) holds for μ as well.

Proof. For a fixed $F \in \mathcal{C}_+$, $U \mapsto \overline{F(\theta U)} F(U)$ is bounded and continuous (depends on finitely many links). By the Portmanteau theorem, the integrals converge: $\int \overline{F(\theta U)} F(U) d\mu_n \rightarrow \int \overline{F(\theta U)} F(U) d\mu$. Since each term is ≥ 0 , the limit is ≥ 0 . See also [10, Ch. 6] and [4, Ch. 2]. Cylinder functions generate a separating subalgebra of $C(X)$ by Stone–Weierstrass, so the quadratic form is determined by its values on $\mathcal{C}_+$. $\square$

Theorem 5.2 (DLR limit, uniqueness, and RP preservation). *For each fixed $a > 0$ there exists a unique infinite-volume Gibbs state μ_a consistent with (3). Moreover, for local f ,*

$$\lim_{\Lambda \uparrow a\mathbb{Z}^4} \mathbb{E}_{\mu_{a,\Lambda}}[f] = \mathbb{E}_{\mu_a}[f], \quad (28)$$

and boundary influences decay exponentially with $\text{dist}(\text{supp} f, \partial\Lambda)$. The state μ_a is reflection positive.

Proof. Define the DLR specification by conditioning on the exterior links. Under either clustering bound (25) or (26), the total influence on a region decays exponentially with distance from its boundary, implying Cauchy convergence of finite-volume expectations and uniqueness (Dobrushin uniqueness). Existence follows by standard compactness. RP of μ_a follows from Lemma 5.1, since each finite-volume measure is RP by Theorem 1.1. $\square$

6 Gauge-invariant local fields and the continuum

For $x \in a\mathbb{Z}^4$ and $\mu < \nu$, define the curvature-energy smear

$$\mathcal{E}_{\mu\nu}^{(a)}(\phi) := \frac{2}{a^4} \sum_{x \in a\mathbb{Z}^4} \phi(x) \left(N - \text{Re tr } U_{p_{\mu\nu}}(x) \right), \quad \phi \in \mathcal{S}(\mathbb{R}^4). \quad (29)$$

Remark 6.1 (Gauge invariance). *Under a gauge transformation $h : x \mapsto h(x) \in G$, links transform as $U_{(x,\mu)} \mapsto h(x) U_{(x,\mu)} h(x + a\hat{\mu})^{-1}$. For a closed plaquette $p_{\mu\nu}(x)$, one has $U_{p_{\mu\nu}}(x) \mapsto h(x) U_{p_{\mu\nu}}(x) h(x)^{-1}$, hence $\text{Re tr}(U_{p_{\mu\nu}}(x))$ and its sum against ϕ are gauge-invariant by Ad-invariance of the trace.*

Lemma 6.2 (Continuum normalization). *For a smooth $\mathfrak{su}(N)$ -valued connection A_μ , the plaquette holonomy satisfies*

$$\text{Re tr } U_{p_{\mu\nu}}(x) = N - \frac{a^4}{2} \text{tr}(F_{\mu\nu}(x)^2) + O(a^6),$$

hence $\mathcal{E}_{\mu\nu}^{(a)}(\phi) \rightarrow \int \phi \frac{1}{2} \text{tr}(F_{\mu\nu}^2)$ as $a \downarrow 0$.

Proof. **Step 1 (Non-abelian Stokes).** The holonomy around the boundary ∂p of a plaquette is

$$U_{p_{\mu\nu}}(x) = \mathcal{P} \exp \left(\oint_{\partial p} A \right),$$

where $\mathcal{P}$ denotes path-ordering. By a non-abelian Stokes expansion (small-area expansion of the ordered exponential), one has

$$\oint_{\partial p} A = \iint_p F_{\mu\nu} dx^\mu \wedge dx^\nu + O(a^3 \|\nabla F\|_{L^\infty(p)}) = a^2 F_{\mu\nu}(x) + O(a^3).$$

Step 2 (Exponential expansion). Since $F_{\mu\nu}(x) \in \mathfrak{su}(N)$ is anti-Hermitian and traceless,

$$U_{p_{\mu\nu}}(x) = \exp(a^2 F_{\mu\nu}(x) + O(a^3)) = \mathbf{1} + a^2 F_{\mu\nu}(x) + \frac{1}{2} a^4 F_{\mu\nu}(x)^2 + O(a^6).$$

Step 3 (Trace and real part). Taking the trace gives

$$\text{tr}(U_{p_{\mu\nu}}(x)) = N + a^2 \text{tr}(F_{\mu\nu}) + \frac{a^4}{2} \text{tr}(F_{\mu\nu}^2) + O(a^6).$$

Because $F_{\mu\nu} \in \mathfrak{su}(N)$ is traceless, $\text{tr}(F_{\mu\nu}) = 0$. Writing $F_{\mu\nu} = iX_{\mu\nu}$ with $X_{\mu\nu}$ Hermitian and traceless, we have

$$\text{tr}(F_{\mu\nu}^2) = \text{tr}((iX_{\mu\nu})^2) = -\text{tr}(X_{\mu\nu}^2) \in \mathbb{R}_{\leq 0}.$$

Therefore, $\text{Re tr}(U_{p_{\mu\nu}}(x)) = N + \frac{a^4}{2} \text{Re tr}(F_{\mu\nu}^2) + O(a^6) = N - \frac{a^4}{2} \text{tr}(X_{\mu\nu}^2) + O(a^6)$, which is exactly the stated formula

$$\text{Re tr } U_{p_{\mu\nu}}(x) = N - \frac{a^4}{2} \text{tr}(F_{\mu\nu}(x)^2) + O(a^6).$$

Step 4 (Riemann sum). Substituting into (29) yields

$$\mathcal{E}_{\mu\nu}^{(a)}(\phi) = \frac{2}{a^4} \sum_{x \in a\mathbb{Z}^4} \phi(x) \left(\frac{a^4}{2} \text{tr}(F_{\mu\nu}(x)^2) \right) + O(a^2) = \sum_{x \in a\mathbb{Z}^4} \phi(x) \text{tr}(F_{\mu\nu}(x)^2) a^4 + O(a^2).$$

As $a \downarrow 0$, the sum converges to $\int_{\mathbb{R}^4} \phi(x) \text{tr}(F_{\mu\nu}(x)^2) dx$, hence to $\int \phi \frac{1}{2} \text{tr}(F_{\mu\nu}^2)$ with our normalization. $\square$

Proposition 6.3 (Uniform exponential moments). *There exists $C < \infty$ such that for all $\lambda \in \mathbb{R}$ and $a > 0$,*

$$\log \mathbb{E}_{\mu_a} \exp\left(\lambda(\mathcal{E}_{\mu\nu}^{(a)}(\phi) - \mathbb{E}_{\mu_a} \mathcal{E}_{\mu\nu}^{(a)}(\phi))\right) \leq \frac{\lambda^2 C \|\phi\|_{L^2}^2}{2}, \quad (30)$$

uniformly in a .

Proof. Differentiate $\Phi(\lambda) = \log \mathbb{E} e^{\lambda F}$ and apply the logarithmic Sobolev inequality to $e^{\lambda F/2}$. The Lipschitz constant of $F = \mathcal{E}_{\mu\nu}^{(a)}(\phi)$ is $\leq C\|\phi\|_{L^2}$ by locality, the chain rule, and compactness of G ; see Appendix . $\square$

Theorem 6.4 (Tightness and RP in the continuum limit). *Let $a_k \downarrow 0$. The n -point functions of finite products of $\mathcal{E}_{\mu\nu}^{(a_k)}(\phi)$ are tight as distributions on $\mathcal{S}(\mathbb{R}^4)^{\times n}$; along a subsequence they converge to limits S_n which are Euclidean-invariant, symmetric, cluster, and reflection-positive.*

Proof. Fix n and a finite set of test functions $(\phi_j)_{j=1}^m$. By Proposition 6.3, each coordinate variable has a sub-Gaussian mgf bound uniform in a , hence uniformly integrable tails. Thus the joint laws in $\mathbb{R}^m$ are tight uniformly in a (Prokhorov). A diagonal selection over a countable dense set $\{\phi_j\}$ in $\mathcal{S}(\mathbb{R}^4)$ produces a subsequence $a_{k_\ell} \downarrow 0$ such that all finite collections converge jointly in distribution, defining multilinear functionals S_n that are Euclidean invariant by translation invariance, symmetric by construction, cluster by dominated convergence and uniform mixing (Sections 4–5), and reflection-positive by Lemma 5.1. $\square$

Theorem 6.5 (OS reconstruction). *The limits $\{S_n\}$ satisfy the OS axioms: Euclidean invariance, symmetry, reflection positivity, cluster, and temperedness. Hence there exists a Hilbert space $\mathcal{H}$, vacuum Ω , a unitary Poincaré representation, and a net of gauge-invariant local fields whose Wightman distributions are the analytic continuations of $\{S_n\}$.*

Proof. Temperedness follows from uniform exponential moments (Schwartz seminorm bounds for smears). The other axioms are verified in Theorem 6.4. The standard OS reconstruction then applies; see Glimm–Jaffe [10], Osterwalder–Schrader [11, 12]. $\square$

7 Uniqueness of the Continuum Limit under Calibration

Fix the calibrated path

$$\alpha(a) = 1 - c a \quad (c > 0 \text{ fixed}), \quad (31)$$

and choose $t_0(a) = t_1(a) = \tau(a)$ so that

$$\alpha(a) = C_D (M_2(\tau(a)) + M_2(\tau(a))) = 2C_D M_2(\tau(a)), \quad (32)$$

cf. (18). For $a > 0$, let μ_a denote the infinite-volume Gibbs state of Theorem 5.2, and let $S_n^{(a)}$ be the n -point Schwinger functions built from the smeared fields of Section 6.

Route B (Dobrushin stability $\Rightarrow$ Cauchy in a)

Lemma 7.1 (Compact t -window and Lipschitz control in t). *There exists a compact $I \subset (0, \infty)$ and $L_I < \infty$ such that $\tau(a) \in I$ for all small a and*

$$\sup_{g \in G} |V_t(g) - V_s(g)| \leq L_I |t - s| \quad \forall s, t \in I. \quad (33)$$

Proof. Continuity and positivity of K_t imply $\inf_g K_t(g) > 0$ on any compact I . Differentiate (1) termwise and use Lemma 2.1. $\square$

Lemma 7.2 (Single-site conditional stability). *Let ν_t be the single-site conditional at time t . Then, for $s, t \in I$,*

$$\sup_{\text{boundary c.f.g.}} \|\nu_t - \nu_s\|_{\text{TV}} \leq 6 L_I |t - s|. \quad (34)$$

Proposition 7.3 (Dobrushin stability for observables). *Let F be a bounded cylinder with support E_F and seminorm $\|\nabla F\|_{\infty; E_F}$. Then for all small a, a' ,*

$$|\mathbb{E}_{\mu_a}[F] - \mathbb{E}_{\mu_{a'}}[F]| \leq \frac{6|E_F|}{1 - \alpha_*} L_I |\tau(a) - \tau(a')| \|\nabla F\|_{\infty; E_F}, \quad \alpha_* := \max\{\alpha(a), \alpha(a')\}. \quad (35)$$

Lemma 7.4 (Calibration transfers to Lipschitz in a). *There exists $C_\tau < \infty$ such that for small a, a' ,*

$$|\tau(a) - \tau(a')| \leq C_\tau c |a - a'|. \quad (36)$$

Corollary 7.5 (Cauchy property). *For every bounded cylinder F ,*

$$|\mathbb{E}_{\mu_a}[F] - \mathbb{E}_{\mu_{a'}}[F]| \leq K_F |a - a'| \quad (37)$$

for some $K_F < \infty$ and all small a, a' .

Theorem 7.6 (Continuum limit exists and is unique (Route B)). *For any finite family of smears (ϕ_j) and any polynomial P in $\{\mathcal{E}_{\mu\nu}^{(a)}(\phi_j)\}$,*

$$\lim_{a \downarrow 0} \mathbb{E}_{\mu_a} [P(\mathcal{E}_{\mu\nu}^{(a)}(\phi_1), \dots, \mathcal{E}_{\mu\nu}^{(a)}(\phi_m))]$$

exists and is unique. Hence $\{S_n^{(a)}\}$ has a unique limit $\{S_n\}$.

Route A (tight + identification via Ward/SD + determinacy)

Lemma 7.7 (Sub-Gaussian control and determinacy). *For any $\phi \in \mathcal{S}(\mathbb{R}^4)$, the centered law of $\mathcal{E}_{\mu\nu}^{(a)}(\phi)$ has a uniform sub-Gaussian mgf (Prop. 6.3); limit laws are moment-determinate. The same holds jointly for finitely many smears.*

Proposition 7.8 (Identification by Ward/SD). *Assume the continuum Ward and Schwinger–Dyson identities hold for any limit point (Section 9). Then all limit points share the same moment family $\{S_n\}$.*

Theorem 7.9 (Continuum limit is unique (Route A)). *Under the hypotheses of Proposition 7.8, all subsequential limits of $\{S_n^{(a)}\}$ coincide.*

8 Nontriviality via Wilson Loops or Connected 4-Point

We present two independent routes to show the continuum limit is non-Gaussian.

Wilson loops and minimal surfaces

Let $C \subset a\mathbb{Z}^4$ be a rectangular loop of size $R \times T$ lying in a coordinate plane; $U(C)$ the ordered product along C ; and $W_C = \text{Re tr } U(C)$. Let $\mathcal{S}(C)$ be the set of plaquette surfaces with boundary C ; $|\Sigma|$ the plaquette number. Let F be the fundamental irrep, with $c_F = c_2(F)$ and $d_F = \dim F$.

Lemma 8.1 (Character insertion per plaquette). $\int_G \chi_F(g) K_t(g) dg = d_F e^{-t c_F}$; hence with $t_{\min} = \min\{t_0, t_1\}$ and $q_F := d_F e^{-t_{\min} c_F} \in (0, 1)$.

Lemma 8.2 (Surface counting). *There exists $\kappa_{\text{surf}} \in (1, \infty)$ s.t. $\#\{\Sigma \in \mathcal{S}(C) : |\Sigma| = A\} \leq \kappa_{\text{surf}}^A$.*

Theorem 8.3 (Area-law upper bound). *For any rectangle C and any volume,*

$$\mathbb{E}_{\mu_a}[W_C] \leq \sum_{\Sigma \in \mathcal{S}(C)} q_F^{|\Sigma|} \leq \sum_{A \geq RT} \kappa_{\text{surf}}^A q_F^A. \quad (38)$$

If $\kappa_{\text{surf}} q_F < 1$, then $\mathbb{E}_{\mu_a}[W_C] \leq \exp(-\sigma_{\text{latt}} RT)$ with $\sigma_{\text{latt}} = -\log(\kappa_{\text{surf}} q_F) > 0$.

Remark 8.4 (Calibration and tension). *Under $\alpha(a) = 1 - ca$ the parameter $\tau(a)$ stays in a compact window I , so $t_{\min}$ is bounded away from 0 and $\sigma_{\text{latt}} > 0$ is a constant. In physical units $A_{\text{phys}} = a^2 RT$, hence $\sigma(a) := \sigma_{\text{latt}}$ satisfies $\sigma(a)/a^2 \rightarrow \infty$; Wilson loops of fixed physical area vanish in the continuum, already excluding any Gaussian/free-field behavior. A finite nonzero continuum tension σ_{cont} would require a different “area calibration” and is not pursued here.*

Connected 4-point survives

Let $Y_p = N - \text{Re tr } U_p$ and $\mathcal{E}_{\mu\nu}^{(a)}(\phi) = \frac{2}{a^4} \sum_x \phi(x) Y_{p\mu\nu}(x)$.

Lemma 8.5 (Local cumulant positivity). *For any compact $I \Subset (0, \infty)$ there exists $c_{\text{loc}}(I) > 0$ such that the single-plaquette fourth cumulant of Y_p under $K_t(g)dg$ satisfies $\kappa_4(Y_p; t) \geq c_{\text{loc}}(I)$ for all $t \in I$.*

Proposition 8.6 (Cumulant stability). *If $\alpha(a) < 1$ then for any finite plaquette set S of bounded diameter and any polynomial $P(\{Y_p\}_{p \in S})$,*

$$|\kappa(P; \mu_a) - \kappa(P; \text{product of single-plaquette laws at } \tau(a))| \leq \frac{C_I}{1 - \alpha(a)} |S|.$$

Theorem 8.7 (Nontrivial connected 4-point). *For $\phi \in \mathcal{S}(\mathbb{R}^4)$, $\phi \not\equiv 0$, there exists $c_4(\phi) > 0$ such that for all small a ,*

$$\kappa_4(\mathcal{E}_{\mu\nu}^{(a)}(\phi)) \geq c_4(\phi).$$

Thus any continuum limit has nonzero fourth cumulant for $\mathcal{E}_{\mu\nu}(\phi)$, and is non-Gaussian.

9 Continuum Ward and Schwinger–Dyson identities

Let $\mu_{a,\Lambda}$ be the HK+HK Gibbs measure (3) on a finite volume Λ , with infinite volume μ_a .

Lattice differential operators

For $e = (x, \mu)$ and $X \in \mathfrak{g}$ define the right-invariant edgewise derivative

$$(\nabla_X^{(e)} f)(U) := \frac{d}{dt} f(\dots, U_e e^{tX}, \dots) \Big|_{t=0}. \quad (39)$$

For $\psi : \mathbb{R}^4 \rightarrow \mathfrak{g}$ smooth, define

$$\mathcal{D}_\psi^{(a)} f := \sum_{e=(x,\mu) \in E(\Lambda)} a^4 \langle \psi_\mu(x), \nabla^{(e)} f \rangle. \quad (40)$$

For a site field $\xi : \mathbb{R}^4 \rightarrow \mathfrak{g}$, define the gauge generator

$$\mathcal{G}_\xi^{(a)} f := \sum_{x \in a\mathbb{Z}^4} \sum_{\mu=1}^4 \left(\langle \xi(x), \nabla^{(x,\mu)} f \rangle - \langle \xi(x), \nabla^{(x-a\hat{\mu},\mu)} f \rangle \right) a^4. \quad (41)$$

Exact lattice Ward and SD identities

Proposition 9.1 (Lattice Ward). *For every cylinder F and site field ξ with finite support, $\mathbb{E}_{\mu_{a,\Lambda}} [\mathcal{G}_\xi^{(a)} F] = 0$.*

Proposition 9.2 (Lattice SD). *For every cylinder F and test field ψ , $\mathbb{E}_{\mu_{a,\Lambda}} [\mathcal{D}_\psi^{(a)} F] = \mathbb{E}_{\mu_{a,\Lambda}} [F \mathcal{D}_\psi^{(a)} S_a]$.*

Continuum identification of the SD insertion

Lemma 9.3 (Continuum SD insertion). *For smooth compactly supported ψ and under the calibration of Section 7, as $a \downarrow 0$,*

$$\mathcal{D}_\psi^{(a)} S_a \longrightarrow \int_{\mathbb{R}^4} \text{tr}(D_\mu F_{\mu\nu}(x) \psi_\nu(x)) dx \quad \text{in } L^1(\mu_{a,\Lambda}) \text{ uniformly on volumes.} \quad (42)$$

Theorem 9.4 (Continuum YM Ward/SD). *For any smooth compactly supported ξ, ψ and polynomial cylinder F in smeared gauge-invariant fields,*

$$\lim_{a \downarrow 0} \mathbb{E}_{\mu_a} [\mathcal{G}_\xi^{(a)} F] = 0, \quad (43)$$

$$\lim_{a \downarrow 0} \mathbb{E}_{\mu_a} [\mathcal{D}_\psi^{(a)} F] = \mathbb{E} \left[F \int_{\mathbb{R}^4} \text{tr}(D_\mu F_{\mu\nu} \psi_\nu) dx \right]. \quad (44)$$

Hence the continuum Schwinger functions satisfy the Yang–Mills Ward/SD system with action density $\frac{1}{2} \text{tr}(F_{\mu\nu}^2)$ (Lemma 6.2).

10 Universality in the Gauge-Invariant Sector under Calibration

We compare the HK+HK model with a Wilson plaquette model tuned to the *same* calibration $\alpha(a) = 1 - ca$. Write μ_a^{HK} for HK+HK and μ_a^{W} for Wilson with anisotropic couplings $\beta_0(a), \beta_1(a)$ chosen so that

$$\alpha^{\text{W}}(a) = 2 \Delta_{\text{deg}} \left(M_2^{\text{W}}(\beta_0(a)) + M_2^{\text{W}}(\beta_1(a)) \right) = 1 - ca = \alpha^{\text{HK}}(a). \quad (45)$$

(Here $M_2^{\text{W}}(\beta) = C_G \beta$ for a group constant $C_G > 0$.)

Let $\mathcal{A}_{\text{GI}}$ be the cylinder algebra generated by finitely many smeared curvature-energy fields.

Route HS+Dob.: quantitative

Define the single-site TV discrepancy

$$\delta(a) := \sup_{\text{boundary cfg.}} \sup_{e \in E(\Lambda)} \left\| (\mathcal{S}_a^{\text{HK}})_e - (\mathcal{S}_a^{\text{W}})_e \right\|_{\text{TV}}. \quad (46)$$

Lemma 10.1 (Holley–Stroock at a site). *If $\nu, \tilde{\nu} \propto e^{-U}, e^{-(U+\delta U)}$ then $\|\nu - \tilde{\nu}\|_{\text{TV}} \leq \text{osc}(\delta U)$.*

Proposition 10.2 (Dobrushin iteration). *If $\alpha^{\text{HK}}(a), \alpha^{\text{W}}(a) \leq \alpha_* < 1$, then for any bounded cylinder F supported on E_F ,*

$$\left| \mathbb{E}_{\mu_a^{\text{HK}}} [F] - \mathbb{E}_{\mu_a^{\text{W}}} [F] \right| \leq \frac{|E_F|}{1 - \alpha_*} \delta(a) \|\nabla F\|_{\infty; E_F}. \quad (47)$$

Definition 10.3 (Character proximity hypothesis). *Let $I \Subset (0, \infty)$. We say HK admits two-character Wilson proximity on I if there exist $\beta_0, \beta_1 \in C(I)$ and $C_I, \eta > 0$ such that for all $t \in I$,*

$$\sup_{g \in G} |V_t(g) - V_{\beta_0(t)}^W(g)| \leq C_I |t - t_*|^{1+\eta}, \quad \sup_{g \in G} |V_t(g) - V_{\beta_1(t)}^W(g)| \leq C_I |t - t_*|^{1+\eta}, \quad (48)$$

for some $t_* \in I$.

Lemma 10.4 (Single-site TV bound). *If (48) holds and $\tau(a) \rightarrow t_*$ linearly (Lemma 7.4), then $\delta(a) = O(a^{1+\eta})$.*

Theorem 10.5 (Universality: HS+Dob.). *Under (45) with $\alpha_* < 1$ and (48), for any polynomial $F \in \mathcal{A}_{\text{GI}}$,*

$$|\mathbb{E}_{\mu_a^{\text{HK}}}[F] - \mathbb{E}_{\mu_a^W}[F]| \leq C(F, I) a^\eta. \quad (49)$$

In particular, the continuum Schwinger functions for the gauge-invariant sector coincide.

Route SD: identification via Ward/SD + renormalization

Definition 10.6 (Renormalization conditions). *Fix smears $\{\phi_j\}_{j=1}^m$ separating linear functionals on $\mathcal{S}(\mathbb{R}^4)$. The two models share RC if*

$$\lim_{a \downarrow 0} \mathbb{E}_{\mu_a^{\text{HK}}}[\mathcal{E}_{\mu\nu}^{(a)}(\phi_j)] = \lim_{a \downarrow 0} \mathbb{E}_{\mu_a^W}[\mathcal{E}_{\mu\nu}^{(a)}(\phi_j)] \quad (1 \leq j \leq m). \quad (50)$$

Proposition 10.7 (SD identification). *If both models satisfy the same continuum Ward/SD identities and RC, then their continuum Schwinger functions coincide on $\mathcal{A}_{\text{GI}}$.*

Theorem 10.8 (Universality: SD route). *Under (45), common Ward/SD identities and RC imply equality of continuum Schwinger functions on $\mathcal{A}_{\text{GI}}$.*

11 Calibrated mass gap from spatial clustering (Route B)

At fixed a , the Route B clustering rate $\mu_B(a)$ gives decay in lattice distance. In *physical* distance $r = a \text{ dist}$, the rate is $\mu_B(a)/a$.

Definition 11.1 (Admissible calibration (Route B)). *A parameter path $t_0(a), t_1(a)$ is admissible if there exists $c_0 > 0$ such that*

$$\lim_{a \downarrow 0} \frac{\mu_B(a)}{a} = c_0 \in (0, \infty). \quad (51)$$

Proposition 11.2 (Existence of calibrated paths). *Let C_D be as in (18) and suppose M_2 satisfies Lemma 2.3. For any $c \in (0, 1)$ there exist continuous $\tau(a)$ such that*

$$\alpha(a) = 2C_D M_2(\tau(a)) = 1 - ca \quad \text{for small } a > 0. \quad (52)$$

Consequently, by (20),

$$\mu_B(a) \geq -\log(1 - ca) \geq ca, \quad (53)$$

and Definition 11.1 holds with $c_0 \geq c$.

Theorem 11.3 (Spectral gap bound). *Under admissible calibration, any continuum limit obtained in Theorem 6.4 has a Hamiltonian spectral gap $\Delta \geq c_0 > 0$ for the class of fields constructed from (29).*

From Euclidean clustering to a Hamiltonian spectral gap

Let S_n be the continuum Schwinger functions. By OS reconstruction (Theorem 6.5), there is a Hilbert space $\mathcal{H}$, vacuum Ω , a self-adjoint Hamiltonian $H \geq 0$, and field operators $\mathcal{E}_{\mu\nu}(\phi, t)$ such that, for $t \geq 0$,

$$S_2^{\text{conn}}(\mathcal{E}_{\mu\nu}(\phi_t), \mathcal{E}_{\mu\nu}(\phi_0)) = \langle \Omega, \mathcal{E}_{\mu\nu}(\phi, 0) e^{-tH} \mathcal{E}_{\mu\nu}(\phi, 0) \Omega \rangle.$$

Lemma 11.4 (Spectral Laplace representation). *For $\Psi := \mathcal{E}_{\mu\nu}(\phi, 0)\Omega - \langle \Omega, \mathcal{E}_{\mu\nu}(\phi, 0)\Omega \rangle \Omega$ one has*

$$\langle \Psi, e^{-tH} \Psi \rangle = \int_{[0, \infty)} e^{-tE} d\mu_\Psi(E),$$

with μ_Ψ the spectral measure of H associated with Ψ .

Theorem 11.5 (Operator-theoretic spectral gap). *Assume there exist $C, m_0 > 0$ such that*

$$|S_2^{\text{conn}}(\mathcal{E}_{\mu\nu}(\phi_x), \mathcal{E}_{\mu\nu}(\phi_0))| \leq C e^{-m_0 |x|} \quad \forall x \in \mathbb{R}^4. \quad (54)$$

Then $\text{spec}(H) \subset \{0\} \cup [m_0, \infty)$ on the cyclic subspace generated by $\mathcal{E}_{\mu\nu}(\phi, 0)\Omega$; in particular H has a spectral gap $\Delta \geq m_0 > 0$ in this sector.

Remark 11.6 (Input of calibrated clustering). *Under Definition 11.1 and (53), we have exponential decay in physical distance with rate $c_0 > 0$ for the centered two-point function of $\mathcal{E}_{\mu\nu}(\phi)$. Euclidean invariance of S_2 yields (54) with $m_0 = c_0$, so Theorem 11.5 implies the Hamiltonian gap $\Delta \geq c_0$.*

Concluding remarks

We provided a purely mathematical construction of HK+HK lattice Yang–Mills with reflection positivity and uniform functional inequalities by two routes, established gradient propagation and two clustering rates (Route A: μ_A fixed; Route B: tunable μ_B), proved infinite-volume uniqueness and continuum OS reconstruction for gauge-invariant fields, derived calibrated mass-gap bounds and upgraded them to an operator-theoretic Hamiltonian gap, and gave universality proofs (quantitative HS+Dob. and structural SD routes).

Appendix A: Uniform LSI via High-Temperature (Route B)

We collect precise high-temperature criteria yielding a volume-uniform logarithmic Sobolev inequality for (2.3).

A.1 Influence coefficients and the Dobrushin constant

Define the influence of e' on e by

$$c(e \leftarrow e') := \sup_{\text{cyl. } f} \frac{\|\nabla_e(\mathbb{E}[f | U_{(E \setminus \{e'\})}])\|_\infty}{\|\nabla_{e'} f\|_\infty}. \quad (55)$$

Set $\alpha := \sup_e \sum_{e' \sim e} c(e \leftarrow e')$. If $\alpha < 1$ then a volume-uniform LSI holds with constant $\rho \geq \rho_{\text{site}}(G, t_0, t_1) (1 - \alpha)$; see Stroock–Zegarlinski [16], Zegarlinski [17], Otto–Reznikoff [13].

A.2 Bounds from local second derivatives (explicit constants)

Let $V_t = -\log K_t$. For a plaquette p touching both e and e' , the mixed second derivative tensor obeys

$$\|\nabla_e \nabla_{e'} V_t(U_p)\| \leq M_2(t), \quad (56)$$

with $M_2(t)$ defined in (8). Since a pair of adjacent links lies in at most two common plaquettes and each link has at most Δ_{deg} neighbours, we obtain

$$c(e \leftarrow e') \leq 2(M_2(t_0) + M_2(t_1)), \quad \alpha \leq 2\Delta_{\text{deg}}(M_2(t_0) + M_2(t_1)) = C_D(M_2(t_0) + M_2(t_1)). \quad (57)$$

A.3 Single-site LSI via Holley–Stroock (explicit lower bound)

Let ν be Haar measure on G with LSI constant $\rho_{\text{Haar}}(G) \geq \rho_G > 0$ (Bakry–Emery on compact manifolds with $\text{Ric} \geq \rho_G$). The single-site conditional measure $\mu_e(\cdot | U_{E \setminus \{e\}})$ has a density proportional to $\exp(-V_{\text{eff}}(U_e))$ with $V_{\text{eff}}(U_e) = \sum_{p \ni e} V_{t_0}(U_p) + \sum_{p \in \mathcal{P}_t: p \ni e} V_{t_1}(U_p)$. Since a link belongs to at most six plaquettes, $\text{osc}(V_{\text{eff}}) \leq 6(\text{osc} V_{t_0} + \text{osc} V_{t_1})$. By Holley–Stroock perturbation, the LSI constant of μ_e is bounded below by

$$\rho_{\text{site}}(G, t_0, t_1) \geq \rho_{\text{Haar}}(G) e^{-\text{osc}(V_{\text{eff}})} \geq \rho_G \exp\left(-6(\text{osc} V_{t_0} + \text{osc} V_{t_1})\right), \quad (58)$$

using $\rho_{\text{Haar}}(G) \geq \rho_G$. The explicit bound on $\text{osc} V_t$ used here is (10).

Appendix B: Lieb–Robinson-Type Velocity Bound (Detailed Route A Proof)

We provide a detailed proof of the gradient propagation bound used in Section 4. The proof iterates a Duhamel inequality, sums over paths on the neighbour graph with multiplicity bounded by Δ_{deg}^n , and uses Stirling to obtain the light-cone form.

B.1 Commutator calculus

For all e, e' ,

$$[\nabla_{e'}, \Delta_e] = 0, \quad (59)$$

$$[\nabla_{e'}, \langle \nabla_e S_a, \nabla_e \rangle] = \begin{cases} \text{Hess}_{e',e}(S_a) \cdot \nabla_e & \text{if } e' \sim e, \\ 0 & \text{otherwise.} \end{cases} \quad (60)$$

B.2 Differential inequality for edgewise gradients

For $g_e(t) := \|\nabla_e P_t^{(a)} f\|_\infty$,

$$\frac{d}{dt} g_{e'}(t) \leq -\kappa_A(a) g_{e'}(t) + J(a) \sum_{e \sim e'} g_e(t), \quad J(a) = 2(M_2(t_0(a)) + M_2(t_1(a))). \quad (61)$$

B.3 Duhamel expansion (one step)

Rewrite (61) as the integral inequality

$$g_{e'}(t) \leq e^{-\kappa_A(a)t} g_{e'}(0) + \int_0^t e^{-\kappa_A(a)(t-s)} J(a) \sum_{e_1 \sim e'} g_{e_1}(s) ds. \quad (62)$$

B.4 Induction to all lengths and path-counting

Assume that for some $n \geq 1$,

$$g_{e'}(t) \leq e^{-\kappa_A t} \sum_e \sum_{\substack{p: e \rightarrow e' \\ |p| \leq n}} \frac{(Jt)^{|p|}}{|p|!} g_e(0) + R_n(t), \quad (63)$$

where R_n collects paths of length $\geq n+1$. Plugging (63) into the RHS of (62) generates paths of length up to $n+1$ with weights $(Jt)^{|p|}/|p|!$; hence (63) holds with $n \rightarrow n+1$. Since the number of length- m paths is $\leq \Delta_{\deg}^m$, we obtain

$$g_{e'}(t) \leq e^{-\kappa_A t} \sum_e \sum_{m \geq d} \frac{(Jt)^m}{m!} \Delta_{\deg}^m g_e(0), \quad d = \text{dist}(e, e'). \quad (64)$$

Stirling $m! \geq (m/e)^m$ and grouping terms yield (23) with $\gamma_A = \frac{1}{2}J$ and $v_A = \Delta_{\deg} J$.

Appendix C: Herbst Inequality from LSI (Proof)

If $\text{Ent}_\mu(h^2) \leq \frac{2}{\rho} \int \|\nabla h\|^2 d\mu$, then for F with Lipschitz constant L one has $\log \mathbb{E} e^{\lambda(F - \mathbb{E}F)} \leq \lambda^2 L^2 / (2\rho)$ by differentiating $\Phi(\lambda) = \log \mathbb{E} e^{\lambda F}$ and applying LSI to $e^{\lambda F/2}$.

Appendix D: Constants at a Glance

- $D = \text{diam}(G)$ (bi-invariant metric). Two-sided HK bounds: (9). Oscillation: (10).
- ρ_G : Ricci lower bound for G in the chosen normalization, cf. (7). Haar LSI: $\rho_{\text{Haar}} \geq \rho_G$ (58).
- $M_i(t)$: global derivatives of $V_t = -\log K_t$ (8). Small- t bounds (11); large- t bounds (12). Constants depend only on geometry $(n, \|\text{Rm}\|_\infty, \text{inj})$.
- $C_D = 2\Delta_{\text{deg}} \leq 24$ (18). Dobrushin contraction: $\alpha(a) \leq C_D(M_2(t_0) + M_2(t_1))$ and $\mu_B \geq -\log \alpha(a)$.
- Route A light-cone parameters: $J = 2(M_2(t_0) + M_2(t_1))$, $v_A = \Delta_{\text{deg}} J$, $\gamma_A = J/2$, clustering rate $\mu_A = 1/(2\Delta_{\text{deg}})$.

General compact simple G : scope and constants

Unless explicitly stated otherwise, all constructions and bounds above extend verbatim to any compact, connected, simple Lie group G equipped with the bi-invariant metric $g = -\text{Killing}$. In particular:

- **Heat kernel and potentials.** The heat kernel admits the class expansion $K_t(g) = \sum_{\lambda \in \widehat{G}} d_\lambda e^{-tc_2(\lambda)} \chi_\lambda(g)$ with strictly positive K_t for all $t > 0$, two-sided Gaussian bounds as in (9), and derivative bounds leading to $M_1(t), M_2(t)$ exactly as in Lemma 2.3; the constants depend only on $(G, \|\text{Rm}\|_\infty, \text{inj}(G))$.
- **Ricci and Casimir.** With $g = -\text{Killing}$, $\text{Ric} = \frac{1}{4}c_2(\text{Ad})g$, hence $\rho_G = \frac{1}{4}c_2(\text{Ad})$; let $c_*(G) := \min_{\lambda \neq 0} c_2(\lambda)$ be the minimal positive quadratic Casimir.
- **Geometry.** Denote by $D(G)$ the diameter and by $C_{\text{Sob}}(G, s)$ the Sobolev embedding constant $H^s(G) \hookrightarrow C^0(G)$ for $s > \frac{\dim G}{2}$; both are finite and explicit in principle.

With these notations, the statements and proofs of: reflection positivity (Thm. 1.1), the two uniform routes (Prop. 2.4 and Appendix), gradient propagation (Prop. 4.2), clustering (Thm. 4.3 and (26)), infinite volume uniqueness (Thm. 5.2), tightness and OS passage (Thm. 6.4), Ward/SD identities (Thm. 9.4), and the calibrated continuum mass scale (Def. 11.1, Thm. 11.3) all hold for general G with constants depending on G only via $(\rho_G, D(G), c_*(G), C_{\text{Sob}}(G, s))$.

Appendix E: Concrete constants for $\text{SU}(N)$ (our normalization)

We fix the bi-invariant metric to be $g = -\text{Killing}$, so that $c_2(\text{Ad}) = 2N$ and $\rho_G = c_2(\text{Ad})/4 = N/2$.

E.1 Quadratic Casimir for the fundamental and the minimum gap

Let F denote the fundamental representation. In this normalization one has

$$c_2(F) = \frac{N^2 - 1}{N}, \quad (65)$$

and $c_2(\lambda) \geq c_2(F)$ for all nontrivial $\lambda \in \widehat{\text{SU}(N)}$; see Fulton–Harris [8, Ch. 13]. Hence

$$c_* = \min_{\lambda \neq 0} c_2(\lambda) = c_2(F) = \frac{N^2 - 1}{N}. \quad (66)$$

E.2 Diameter of $\text{SU}(N)$

For the metric $g = -\text{Killing} = 2N(-\text{Tr})$,

$$D^2 = \begin{cases} 2N^2 \pi^2, & N \text{ even,} \\ 2N(N-1) \pi^2, & N \text{ odd.} \end{cases} \quad (67)$$

E.3 Explicit numerical instantiation

For $\text{SU}(2)$: $c_2(\text{Ad}) = 4$, $\rho_G = 1$, $c_2(F) = \frac{3}{2}$, and $D^2 = 8\pi^2$. For $\text{SU}(3)$: $c_2(\text{Ad}) = 6$, $\rho_G = \frac{3}{2}$, $c_2(F) = \frac{8}{3}$, and $D^2 = 12\pi^2$.

Appendix F: Explicit two-character Wilson approximations on $\text{SU}(2)$ and $\text{SU}(3)$

We construct, for $G = \text{SU}(2)$ and $G = \text{SU}(3)$, explicit coefficient functions $\beta_F(t), \beta_A(t) \in C^1(I)$ and $c(t) \in C^1(I)$ on a compact interval $I \Subset (0, \infty)$ such that

$$\sup_{g \in G} \left| V_t(g) - c(t) - \beta_F(t)(d_F - \text{Re } \chi_F(g)) - \beta_A(t)(d_A - \text{Re } \chi_A(g)) \right| \leq C_I |t - t_*|^2, \quad (68)$$

for some $t_* \in I$, with an explicit C_I . This realizes (48) with $\eta = 1$. The construction uses an H^s -orthogonal projection onto $\text{span}\{1, \phi_2, \phi_3\}$ with $s > \frac{\dim G}{2}$ and Sobolev embedding constants $C_{\text{Sob}}(G, s)$; see Section F.1–F.3 for details and constants. (We omit the repetition here for brevity; see the extended Appendix F text in the prior version.)

Appendix G: Numerical upper bounds for $\mathcal{M}_{s,2}(I)$ on $\text{SU}(2)$ and $\text{SU}(3)$

We implement a truncate-and-tail strategy to bound $\mathcal{M}_{s,2}(I)$ (defined in (69) below) on a window $I = [t_{\min}, t_{\max}]$, using exact representation data for both groups. (See the extended Appendix G text for the full derivation and worked examples.)

G.1 Definition and prefactor

$$\mathcal{M}_{s,2}(I) := \sup_{t \in I} \frac{1}{\left(\inf_{g \in G} K_t(g)\right)^2} \sum_{\lambda \neq 0} d_\lambda^2 c_2(\lambda)^2 (1 + c_2(\lambda))^{s/2} e^{-t c_2(\lambda)}. \quad (69)$$

With D^2 as in Appendix and $n = \dim G$, the two-sided Gaussian bound yields

$$\sup_{t \in I} \frac{1}{\left(\inf_g K_t\right)^2} \leq \frac{t_{\min}^n}{C_-^2} \exp\left(\frac{D^2}{2 t_{\min}}\right). \quad (70)$$

Appendix H: Fully numeric $B_-(I)$ and $\mathcal{M}_{s,2}(I)$ on $I = [0.8, 1.2]$

Using direct evaluation of the exact representation series, we obtain effective $C_-^{\text{eff}}(I; G)$ and thus fully numeric $B_-(I)$ and $\mathcal{M}_{s,2}(I)$ on $I = [0.8, 1.2]$. For $\text{SU}(2)$: $C_-^{\text{eff}} = 1.7856 \times 10^7$, $B_-(I) = 4.3381 \times 10^6$, and $\mathcal{M}_{3,2}(I) \leq 6.2381 \times 10^8$. For $\text{SU}(3)$: $C_-^{\text{eff}} = 7.2096 \times 10^{10}$, $B_-(I) = 4.5317 \times 10^9$, and $\mathcal{M}_{5,2}(I) \leq 5.7583 \times 10^{13}$. (See the extended Appendix H for computational details.)

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